Deva Anand

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Education

University of Massachusetts Amherst — **September 2025 – May 2027 (Expected)**
Master of Science in Computer Science — Amherst, MA
Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies — **August 2019 – May 2023**
Bachelor of Engineering in Computer Science and Engineering — Chennai, India

Skills

Languages	Python, C++, Java, JavaScript, TypeScript, SQL, Bash
Data & Pipelines	Apache Spark / PySpark, high-throughput ETL, data aggregation, distributed-data processing, real-time (WebSocket)
AI & Agents	LLM agents, agentic workflows, RAG, reranking, OpenAI / Anthropic / Groq APIs, prompt design, LLM eval
Machine Learning	PyTorch, scikit-learn, supervised / unsupervised learning, classification, NumPy, pandas
Cloud & Security	AWS (S3, KMS), Docker, GCP, Azure, Git, CI/CD, encryption, authentication, VAPT remediation

Experience

Wysa — **July 2023 – July 2025**
Backend Engineer — Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability across releases and remediated **20+ VAPT findings**.
- Implemented **AWS KMS envelope encryption** for PII / clinical data on the multi-tenant stack and enforced a **90-day data-retention** policy, privacy-first handling of sensitive data.

Cario Growth Services Pvt. Ltd. — **November 2022 – May 2023**
Backend Developer Intern — Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by PostgreSQL, and integrated LLM-powered context-aware replies into a chat product, iterating on prompt design and response-quality heuristics for a real customer-facing experience.

Projects

Spark ETL - High-Throughput Data Pipeline — github.com/Deva-1903/Spark-ETL-Optimization
Apache Spark / PySpark, SQL, Python | 1.4B-row dataset

- Built and tuned a high-throughput data pipeline over a **1.4B-row** dataset, cutting end-to-end runtime **25%** via broadcast joins, adaptive query execution, and partition tuning, and reducing tail-latency ratio (P99/P50) from **2.14x to 1.74x** via pruning and filter pushdown, profiling with Spark UI.

AgenticSearch - Agentic LLM Workflow — github.com/Deva-1903/ciir_agentic_search
Python, FastAPI, OpenAI / Groq, SQLite

- Built an **agentic LLM workflow** that plans typed retrieval, tool use, deterministic extraction with LLM fallback, and output verification, returning **cell-level provenance** for every value.
- Hardened the service with **150+ automated tests**, provider routing/fallback, and pipeline instrumentation, the kind of reliability real-time decisioning needs.

KG2RAG-Enhanced - RAG & Multi-Hop Retrieval — github.com/Deva-1903/KG2RAG-685-NLP
Python, Ollama, sentence-transformers, PyTorch | UMass CS 685 Advanced NLP (team)

- Led the **multi-view retrieval** component of a team **RAG** pipeline (sub-question decomposition, RRF fusion + cross-encoder reranking + MMR), lifting supporting-fact recall **+2.68%** over the KG^2RAG baseline on a 4,905-question evaluation.

Deep Learning Models (CS 689) — github.com/Deva-1903/UMASS_CICS_689
PyTorch, JAX, NumPy | UMass CS 689 Advanced ML

- Trained and benchmarked **5 deep architectures** (CNN, ResNet) for **classification** on CIFAR-10 across 3 optimizers, and implemented reverse-mode autodiff from scratch (validated against JAX).
- Implemented generative models from scratch in JAX (a normalizing flow and a DDPM), **unsupervised** density modeling, deriving the ELBO via variational inference.

Publication & Extracurriculars

- **Publication:** *"Alzheimer's Disease Classification using Transfer Learning,"* **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.